Install Required Libraries

```
!pip install langchain langchain-community langchain-core
!pip install chromadb
!pip install pypdf
!pip install sentence-transformers
!pip install groq
!pip install langgraph
!pip install langchain-community
!pip install langchain-text-splitters
```

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Requirement already satisfied: urllib3<3,>=1.26 in /usr/local/lib/python3.12/dist-packages (from requests>=2.0.
```

Import Libraries

```
import os

from langchain_community.document_loaders import PyPDFLoader
from langchain_text_splitters import RecursiveCharacterTextSplitter
from langchain_community.vectorstores import Chroma
from langchain_community.embeddings import HuggingFaceEmbeddings

from groq import Groq
```

Set Groq API Key

```
os.environ["GROQ_API_KEY"] = "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"

client = Groq(api_key=os.environ["GROQ_API_KEY"])
```

Upload PDF

```
from google.colab import files
uploaded = files.upload()
```

Choose files sample.pdf

sample.pdf(application/pdf) - 218277 bytes, last modified: 22/04/2026 - 100% done

```
Saving sample.pdf to sample (1).pdf
```

Load the PDF

```
loader = PyPDFLoader("sample.pdf")
documents = loader.load()

print("Total pages:", len(documents))
```

```
Total pages: 3
```

Chunk the Document

```
text_splitter = RecursiveCharacterTextSplitter(
    chunk_size=500,        # size of each chunk
    chunk_overlap=50       # overlap for better context
)

chunks = text_splitter.split_documents(documents)

print("Total chunks:", len(chunks))
print("Sample chunk:\n", chunks[0].page_content)
```

```
Total chunks: 12
Sample chunk:
 Frequently Asked Questions (FAQs)

Q1. Can the Card be used immediately after it is purchased?
Answer- Yes, your State Bank Vishwa Yatra  Foreign Travel Card can be used
immediately after purchase except in India, Nepal and Bhutan.

Q2. Can I use the Add-on cards simultaneously?
Answer- Yes, Add-on cards can be used simultaneously along with the original card.

Q3. Are there any expenses for which this Card may not be used?
```

Create Embeddings

```
embedding_model = HuggingFaceEmbeddings(
    model_name="sentence-transformers/all-MiniLM-L6-v2"
)
```

```
/tmp/ipykernel_1274/2127729888.py:1: LangChainDeprecationWarning: The class `HuggingFaceEmbeddings` was deprecate
  embedding_model = HuggingFaceEmbeddings(
/usr/local/lib/python3.12/dist-packages/huggingface_hub/utils/_auth.py:93: UserWarning:
The secret `HF_TOKEN` does not exist in your Colab secrets.
To authenticate with the Hugging Face Hub, create a token in your settings tab (https://huggingface.co/settings/t
You will be able to reuse this secret in all of your notebooks.
Please note that authentication is recommended but still optional to access public models or datasets.
  warnings.warn(
Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN to enable higher rate limi
WARNING:huggingface_hub.utils._http:Warning: You are sending unauthenticated requests to the HF Hub. Please set a
modules.json: 100%  349/349 [00:00<00:00, 37.7kB/s]
config_sentence_transformers.json: 100%  116/116 [00:00<00:00, 10.7kB/s]
README.md:  10.5k/? [00:00<00:00, 978kB/s]
sentence_bert_config.json: 100%  53.0/53.0 [00:00<00:00, 3.22kB/s]
config.json: 100%  612/612 [00:00<00:00, 49.2kB/s]
model.safetensors: 100%  90.9M/90.9M [00:01<00:00, 458MB/s]
Loading weights: 100%  103/103 [00:00<00:00, 457.64it/s, Materializing param=pooler.dense.weight]
BertModel LOAD REPORT from: sentence-transformers/all-MiniLM-L6-v2
Key                     | Status     |  |
------------------------+------------+--+-
embeddings.position_ids | UNEXPECTED |  |

Notes:
- UNEXPECTED   :can be ignored when loading from different task/architecture; not ok if you expect identical arch
tokenizer_config.json: 100%  350/350 [00:00<00:00, 34.3kB/s]
vocab.txt:  232k/? [00:00<00:00, 2.06MB/s]
tokenizer.json:  466k/? [00:00<00:00, 7.20MB/s]
special_tokens_map.json: 100%  112/112 [00:00<00:00, 12.9kB/s]
config.json: 100%  190/190 [00:00<00:00, 13.3kB/s]
```

Store in ChromaDB

```
vectorstore = Chroma.from_documents(
    documents=chunks,
    embedding=embedding_model,
    persist_directory="./chroma_db"
)

print("Vector DB created successfully")
```

```
Vector DB created successfully
```

Create Retriever

```
retriever = vectorstore.as_retriever(search_kwargs={"k": 3})
```

Test Retrieval

```
query = "Can I use the card for online transactions?"

docs = retriever.invoke(query)

for i, doc in enumerate(docs):
    print(f"\n--- Result {i+1} ---")
    print(doc.page_content)
```

```
--- Result 1 ---
prepaid.onlinesbi.com website and using your 16 digit card number and 6 digit
password.

Q13. Can I use the Card for on-line transactions?
Answer: Presently, the Card is not enabled for e-Commerce.
Q14. Do I need to have account with State Bank to purchase the Vishwa
Yatra Foreign Travel Card?
Answer: No, you need not be a State Bank account holder for purchasing State Bank
Vishwa Yatra Foreign Travel Card.
Q15. Can re-loading of the card by a family member or relatives possible?

--- Result 2 ---
Frequently Asked Questions (FAQs)
```

```
Q1. Can the Card be used immediately after it is purchased?
Answer- Yes, your State Bank Vishwa Yatra  Foreign Travel Card can be used
immediately after purchase except in India, Nepal and Bhutan.

Q2. Can I use the Add-on cards simultaneously?
Answer- Yes, Add-on cards can be used simultaneously along with the original card.

Q3. Are there any expenses for which this Card may not be used?

--- Result 3 ---
Answer- All transactions related to gambling or any unlawful activity are prohibited
onto the card.
Note: Please note that for any temporary charges like deposits at hotels, on hiring of
cars, the Bank will debit the  card for any such amount. If any reversal of amount is
required; it will be reversed and credited to the Card Account within a maximum period
of 60 days.

Q4.Can the value on the Card be topped up from any authorised Branch?
```

Create LLM Function

```
def generate_answer(query, context):
    prompt = f"""
You are a helpful customer support assistant.

Answer the question based ONLY on the context below.
If the answer is not in the context, say "I don't know."

Context:
{context}

Question:
{query}

Answer:
"""

    response = client.chat.completions.create(
        model="llama3-8b-8192",  # fast + free in Groq
        messages=[{"role": "user", "content": prompt}],
        temperature=0
    )

    return response.choices[0].message.content
```

Combine Retrieval + LLM

```
def rag_pipeline(query):
    docs = retriever.invoke(query)

    context = "\n\n".join([doc.page_content for doc in docs])

    answer = generate_answer(query, context)

    return answer
```

Test Chatbot

```
def generate_answer(query, context):
    prompt = f"""
You are a helpful customer support assistant.

Answer the question based ONLY on the context below.
If the answer is not in the context, say "I don't know."

Context:
{context}

Question:
{query}

Answer:
"""

    response = client.chat.completions.create(
        model="llama-3.1-8b-instant",
        messages=[{"role": "user", "content": prompt}],
        temperature=0
    )
```

```
    return response.choices[0].message.content
```

```
!pip install langgraph
```

```
Requirement already satisfied: langgraph in /usr/local/lib/python3.12/dist-packages (1.1.6)
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Requirement already satisfied: urllib3<3,>=1.26 in /usr/local/lib/python3.12/dist-packages (from requests>=2.0.0-
```

Define State

```
from typing import TypedDict

class GraphState(TypedDict):
    query: str
    context: str
    answer: str
    confidence: float
```

Processing Node

```
def processing_node(state):
    query = state["query"]

    docs = retriever.invoke(query)
    context = "\n\n".join([doc.page_content for doc in docs])

    answer = generate_answer(query, context)

    # Simple confidence logic
    if "I don't know" in answer:
        confidence = 0.2
    else:
        confidence = 0.9

    return {
        "context": context,
        "answer": answer,
        "confidence": confidence
    }
```

Decision Function

```
def route_decision(state):
    if state["confidence"] < 0.5:
        return "hitl"
    else:
        return "output"
```

Output Node

```
def output_node(state):
    print("\n✅ Final Answer:\n", state["answer"])
    return state
```

HITL Node

```
def hitl_node(state):
    print("\n⚠️ Escalating to Human Support...")
    print("Query:", state["query"])

    human_response = input("👮 Enter human response: ")

    return {
        "answer": human_response,
        "confidence": 1.0
    }
```

Build LangGraph

```
from langgraph.graph import StateGraph

builder = StateGraph(GraphState)

builder.add_node("process", processing_node)
builder.add_node("output", output_node)
builder.add_node("hitl", hitl_node)

builder.set_entry_point("process")

builder.add_conditional_edges(
    "process",
    route_decision,
    {
        "output": "output",
        "hitl": "hitl"
    }
)

builder.set_finish_point("output")

graph = builder.compile()
```

Run the Graph

```
query = "What is my account balance?"

graph.invoke({"query": query})
```

```
⚠️ Escalating to Human Support...
Query: What is my account balance?
👮 Enter human response: 450
{'query': 'What is my account balance?',
 'context': "made by account payee banker's cheque/ draft/ credit to account whereas if it is below \nRs. 50,000
Cash disbursement is also available.\n\nQ9. Do the ATM receipts show the amount withdrawn and the balance
\navailable?  \nAnswer- Depending on the capability of the ATMs, the amount withdrawn and/or \nbalance available
may be shown.  \n \nQ.10. In countries where ATM instructions are in an unfamiliar language, \nwhom can I
approach for assistance?  \nAnswer- English is available at most VISA ATMs.  \n \nQ11. Is there a fee for
obtaining refunds on the balance on the State Bank \nVishwa Yatra Foreign Travel Card?\n\nYou can also request
for partial refund.  It may be noted that customer/cardholder is \nbound to RBI/FEMA regulations to retain the
foreign currency up to the permissible limit \nin the card. \nPlease Note: \nIf the balance in the card is
claimed after expiry of the card, the balance will be \nrefunded after deduction of USD -5/GBP-5/Euro-5/AUD-
5/CAD-5/JPY-500/SAR-20/   \nSGD-6. \nFurther, if the balance in the card is below USD -5/GBP-5/Euro-5/AUD-5/CAD-
5/JPY-",
 'answer': '450',
 'confidence': 1.0}
```